THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 4: **Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $(2x + 3y)^2 =$ 1

(a) $4x^2 + 12xy + 9y^2$ **(b)** $4x^2 + 6xy + 9y^2$ **(c)** $4x^2 - 12xy + 9y^2$ **(d)** $4x^2 + 9y^2$

2. $(3x - 2y)^2 =$ 1

(a) $9x^2 - 12xy + 4y^2$ **(b)** $9x^2 + 4y^2$ **(c)** $9x^2 - 6xy + 4y^2$ **(d)** $9x^2 + 12xy + 4y^2$

3. $(x + 7)(x - 3) =$ 1

(a) $x^2 + 10x - 21$ **(b)** $x^2 - 4x - 21$ **(c)** $x^2 + 4x + 21$ **(d)** $x^2 + 4x - 21$

4. By identity, $103^2 =$ 1

(a) 10608 **(b)** 10609 **(c)** 10611 **(d)** 10610

5. $x^2 + 8x + 16$ is the square of: 1

(a) $(x - 4)$ **(b)** $(x + 4)$ **(c)** $(x + 8)$ **(d)** $(x + 16)$

6. Factorise $x^2 + 7x + 10$: 1

(a) $(x - 2)(x - 5)$ **(b)** $(x + 4)(x + 5)$ **(c)** $(x + 2)(x + 5)$ **(d)** $(x + 3)(x + 4)$

7. If $x + y = 9$ and $xy = 14$, then $x^2 + y^2 =$ 1

(a) 52 **(b)** 53 **(c)** 55 **(d)** 54

8. $(a + b)^3$ equals: 1

(a) $a^3 + b^3$ **(b)** $a^3 + 3a^2b - 3ab^2 + b^3$ **(c)** $a^3 + 3ab + b^3$ **(d)** $a^3 + 3a^2b + 3ab^2 + b^3$

9. $(3x + 4y)^2 =$ 1

(a) $9x^2 - 24xy + 16y^2$ **(b)** $9x^2 + 12xy + 16y^2$ **(c)** $9x^2 + 24xy + 16y^2$ **(d)** $9x^2 + 16y^2$

10. $(2x - 5y)^2 =$ 1

(a) $4x^2 - 20xy + 25y^2$ **(b)** $4x^2 - 10xy + 25y^2$ **(c)** $4x^2 + 25y^2$ **(d)** $4x^2 + 20xy + 25y^2$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Expand $(3x + 5y)^2$. 2

12. Find 104^2 using an identity. 2

13. Expand $(x + 5)(x + 8)$. 2

14. Expand $(2x - 7y)^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Expand $(2x + 3y)^2$ and verify for $x = 1$, $y = 1$. 3

16. Factorise $x^2 + 8x + 12$. 3

17. Evaluate 103×107 using $(x + a)(x + b)$. 3

OR

Expand $(3x - 4y)^2$ and verify for $x = 1$, $y = 1$. 3

18. Factorise $x^2 - 7x + 10$. 3

OR

Evaluate 97×95 using $(x + a)(x + b)$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Expand $(2x - 3y)^2$ and find its value when $x = 3$ and $y = 1$. 5

OR

Expand $(x + 2)^3$. 5

20. Expand $(3x - y)^2$ and find its value when $x = 2$ and $y = 4$. 5

OR

Expand $(2x + 1)^3$. 5

END OF QUESTION PAPER — Total: 40 Marks

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